

Atoms and Molecules

AI Tutor - Chemistry - Lesson 2

The Atom

An atom is the smallest unit of an element that retains its chemical properties. Atoms consist of a nucleus (containing protons and neutrons) surrounded by electrons.

- **Proton** - positively charged, found in the nucleus.
- **Neutron** - no charge, found in the nucleus.
- **Electron** - negatively charged, orbits the nucleus.

Elements and the Periodic Table

An element is a substance made of only one type of atom (e.g. Oxygen, Hydrogen, Carbon). The periodic table organizes all known elements by their atomic number and chemical properties.

Molecules

A molecule forms when two or more atoms bond together. For example, water (H_2O) is a molecule made of two hydrogen atoms bonded to one oxygen atom.